

Tubular, Interstitial, Stone, Obstructive, and GU Disorders

This chapter completes the renal pathology block by moving from glomerular patterns to tubular injury, interstitial inflammation, papillary necrosis, renal tubular acidosis, stones, urinary obstruction, infection, cystic disease, and selected genitourinary syndromes. The goal is not to memorize isolated names, but to build a usable illness-script map: where the injury occurs, what appears in the urine, what threatens kidney function, and when the answer is urgent drainage, drug withdrawal, antibiotics, stone management, or supportive care.

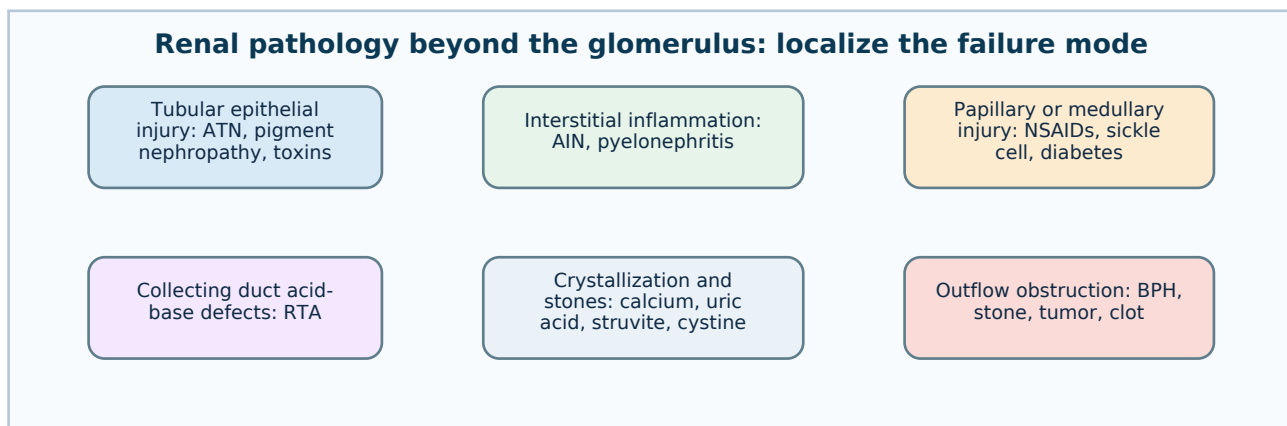
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Why this matters clinically

Tubular and obstructive renal disease is where renal medicine becomes time-sensitive. Glomerular disease often asks, "Is the urine nephritic or nephrotic?" Tubular and interstitial disease asks, "Can the nephron still concentrate urine, reclaim sodium, excrete acid, and avoid back-pressure?" A patient with muddy brown granular casts after shock, sterile pyuria after a new antibiotic, flank pain with hydronephrosis, or fever with an obstructing stone needs a different response than a patient with albumin-dominant proteinuria. In this chapter, the most important clinical split is reversible functional stress versus structural tubular injury versus inflammation of the interstitium versus blocked outflow.

The uploaded renal disease pages emphasize the same practical structure: first exclude prerenal and postrenal causes, then use urinalysis, urine chemistry, renal ultrasound, and medication review to decide whether the remaining pattern is tubular necrosis, glomerular disease, interstitial disease, or obstruction. The AKI figure and urinalysis table in the source section anchor this logic: bland sediment points away from intrinsic cellular damage; granular casts suggest tubular injury; RBC casts suggest glomerular inflammation; WBC casts suggest pyelonephritis or interstitial nephritis; ultrasound is used to look for obstruction.

Core localization map



This localization map is the chapter scaffold. Tubular epithelial injury produces impaired reabsorption and concentration, often with granular casts and a FeNa that no longer looks avidly sodium-retentive. Interstitial inflammation adds pyuria, WBC casts, fever/rash/eosinophilia in drug-triggered cases, and sometimes sterile urine cultures. Medullary/papillary disease can cause hematuria, colic-like pain, sloughed papillae, and concentrating defects. Stones and obstruction injure the kidney by increasing intratubular pressure, triggering

infection risk, and converting a focal urinary tract problem into AKI when both kidneys, a solitary kidney, or the bladder outlet are involved.

Acute tubular necrosis and tubular injury

Acute tubular necrosis is the prototype intrinsic AKI. It usually follows ischemia or nephrotoxins. Ischemic ATN develops after sustained hypoperfusion: shock, major hemorrhage, sepsis, severe heart failure, or prolonged prerenal azotemia. Nephrotoxic ATN develops when tubular cells are injured by substances such as aminoglycosides, amphotericin B, cisplatin, iodinated contrast in susceptible patients, radiocontrast-associated hemodynamic stress, or endogenous pigments from rhabdomyolysis and hemolysis. The central concept is loss of tubular epithelial function: sodium is no longer avidly reclaimed, urine cannot be concentrated well, dead cells and protein form granular casts, and GFR falls from vasoconstriction, tubular obstruction, and backleak of filtrate.

Clinically, ATN often evolves through an initiating insult, an oliguric or maintenance phase, and a recovery or diuretic phase. The oliguric phase may last days to weeks and is dangerous because potassium, acid, phosphate, sodium, water, and nitrogenous wastes accumulate. The recovery phase can produce brisk urine output before tubular concentrating ability has fully returned, so patients can become hypovolemic or develop electrolyte wasting even while creatinine is improving. The uploaded source highlights this phase logic and the need to monitor daily weights, input/output, electrolytes, BUN, creatinine, and acid-base status in AKI.

Feature	Prerenal physiology	ATN / intrinsic tubular injury	Practical interpretation
Trigger	Low effective arterial blood volume: dehydration, heart failure, cirrhosis, sepsis	Prolonged ischemia or nephrotoxin/pigment exposure	Prerenal disease can become ATN if hypoperfusion persists.
Urine sediment	Bland or hyaline casts	Muddy brown granular casts; renal tubular epithelial cells	Sediment often gives the fastest clue when chemistry is confounded.
BUN:Cr ratio	Often >20:1	Often closer to 10-15:1	Not reliable alone, but helps when paired with story and urine findings.
Urine sodium / FeNa	Low urine Na; FeNa often <1%	Urine Na often >40 mEq/L; FeNa often >2%	Diuretics can falsely raise urine Na; use FeUrea or sediment if needed.
Urine osmolality	Concentrated, often >500 mOsm/kg	Less concentrated, often 250-350 mOsm/kg	Tubular injury means the kidney cannot conserve water/sodium normally.
Treatment emphasis	Restore perfusion and stop hemodynamic insults	Supportive care, stop toxin, manage complications, dialysis if needed	No magic reversal: prevent progression and treat life threats.

High-yield trap: FeNa is a tool, not a verdict. Recent loop/thiazide use, CKD, contrast exposure, pigment nephropathy, and sepsis can blur classic thresholds. In a patient on diuretics, FeUrea <35% supports prerenal physiology, but urine microscopy and the timeline still matter.

Pigment nephropathy and rhabdomyolysis

Rhabdomyolysis injures kidney tubules through myoglobin-mediated toxicity, intratubular cast formation, renal vasoconstriction, volume depletion from third spacing into injured muscle, and electrolyte shifts. Causes include crush injury, prolonged immobilization, seizures, exertional heat illness, statin toxicity, alcohol or stimulant intoxication, and metabolic muscle disease. Clues include muscle pain or swelling, dark urine, urine dipstick positive for blood with few or no RBCs on microscopy, very high creatine kinase, hyperkalemia, hyperphosphatemia, hypocalcemia early, and possible hypercalcemia during recovery.

Management is early isotonic crystalloid when the patient can tolerate volume, aggressive monitoring of potassium and calcium, avoidance of nephrotoxins, and treatment of compartment syndrome or the underlying trigger. Sodium bicarbonate and mannitol are sometimes discussed, but routine use is not the core intervention. The clinical anchor is renal perfusion plus complication control: severe hyperkalemia, refractory acidosis, volume overload, or uremic complications may require dialysis.

Acute interstitial nephritis

Acute interstitial nephritis is inflammation of the renal interstitium and tubules. The classic board-style triad is fever, rash, and eosinophilia after a medication exposure, but real patients often present with only AKI, sterile pyuria, WBC casts, mild proteinuria, and a medication list that changed days to weeks earlier. Common triggers include beta-lactam antibiotics, sulfonamides, rifampin, fluoroquinolones, proton pump inhibitors, NSAIDs, immune checkpoint inhibitors, and some diuretics. Infections and systemic diseases such as sarcoidosis, Sjogren disease, IgG4-related disease, and tubulointerstitial nephritis with uveitis can also cause interstitial nephritis.

The pathophysiology is immune-mediated injury around tubules rather than primary glomerular capillary inflammation. That is why hematuria is usually less dramatic than in nephritic syndromes, proteinuria is often subnephrotic, and WBC casts matter. NSAID-associated AIN can be atypical because it may cause heavier proteinuria through a minimal-change-like podocyte injury. The first step is to stop the offending agent. Kidney biopsy is considered when the diagnosis is uncertain, kidney function fails to improve, or steroid treatment is being weighed. Corticosteroids are sometimes used, especially when early recognition and persistent dysfunction suggest ongoing inflammation, but the key test-taking and clinical move is still drug withdrawal plus supportive renal care.

Pattern	Most likely process	Urine/lab clue	Action
New antibiotic or PPI, AKI, sterile pyuria	Drug-induced AIN	WBC casts, eosinophilia sometimes, mild proteinuria	Stop drug; review all OTC meds; consider nephrology/biopsy if no improvement.
NSAID exposure, AKI, edema, heavier proteinuria	NSAID-associated interstitial + podocyte injury	Protein may be more prominent than classic AIN	Stop NSAID; evaluate for nephrotic features if present.
Flank pain, fever, pyuria, positive culture	Pyelonephritis	WBC casts can occur; bacteriuria	Antibiotics; image if severe, recurrent, obstructed, or not improving.
Sterile pyuria, uveitis, systemic features	Systemic tubulointerstitial disease	Inflammatory markers, extrarenal clues	Search for sarcoid, Sjogren, IgG4, TINU, autoimmune disease.

Papillary necrosis and medullary vulnerability

The renal papilla is vulnerable because the medulla lives at low oxygen tension and high solute concentration. Papillary necrosis occurs when medullary blood supply or tubular integrity is impaired. Classic associations can be remembered as "NSAIDs, sickle cell, diabetes, obstruction, pyelonephritis, analgesic overuse"; other causes include renal vein thrombosis, transplant rejection, and severe dehydration. Patients may present with gross or microscopic hematuria, flank pain, colic from sloughed papillae, sterile pyuria, infection, or obstructive AKI if papillary fragments block urine flow.

Sickle cell trait is an important exam clue: even trait, not only sickle cell disease, can cause renal medullary ischemia because the hypertonic and hypoxic medulla promotes sickling. Consequences include papillary necrosis, painless hematuria, hyposthenuria, and impaired urinary concentration. NSAIDs worsen medullary ischemia by inhibiting prostaglandin-mediated afferent arteriolar dilation, especially when renal perfusion is already stressed.

Renal tubular acidosis as a tubular function disorder

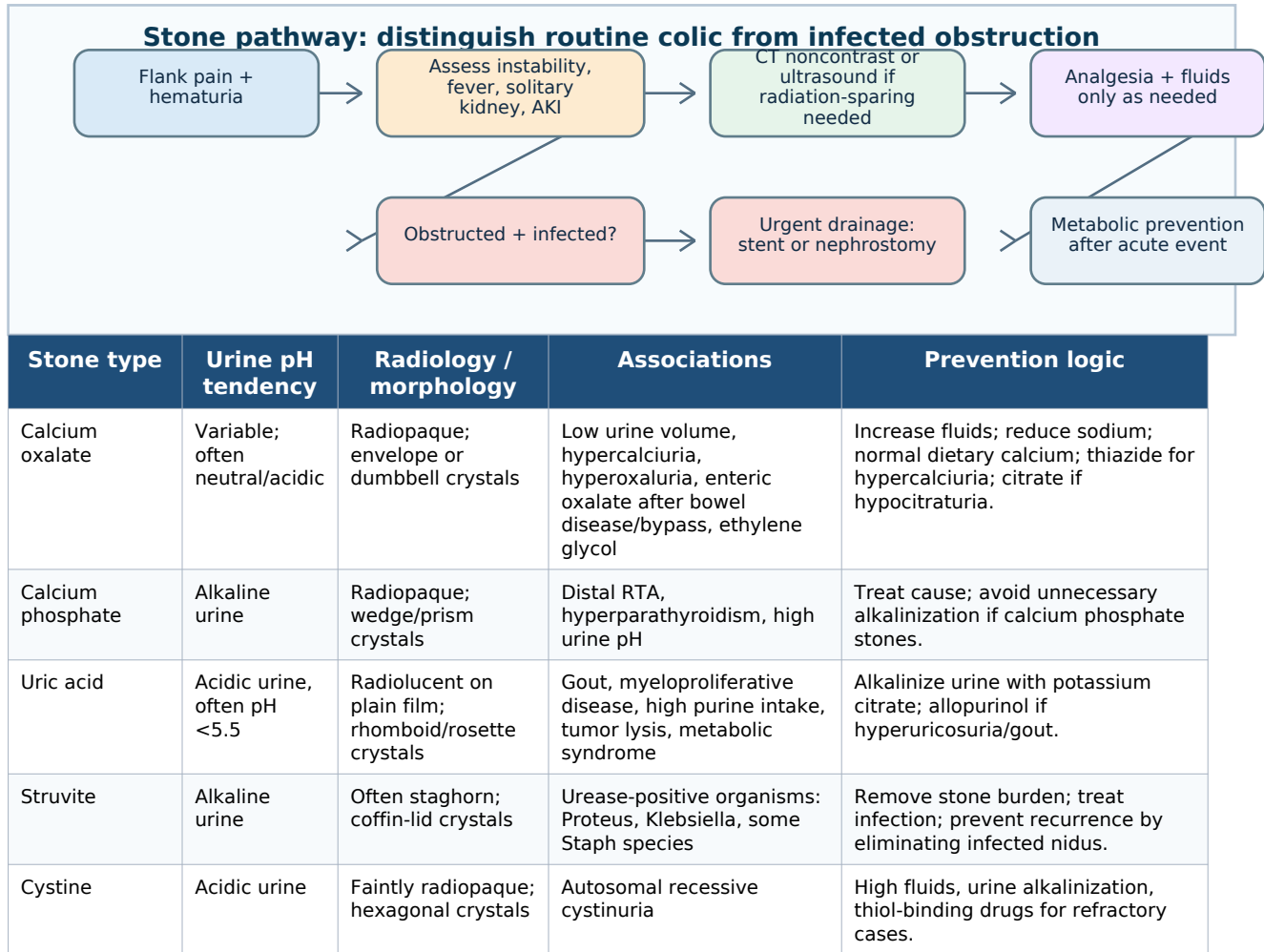
Renal tubular acidosis is best viewed as a failure of renal acid handling that produces a normal anion gap metabolic acidosis. The glomerulus may filter adequately, but the tubules fail at one of three acid-base tasks: distal hydrogen secretion, proximal bicarbonate reclamation, or aldosterone-mediated sodium reabsorption/potassium-hydrogen secretion. RTA was introduced in the acid-base chapter, but it belongs here anatomically because each subtype localizes to a tubular segment and predicts potassium direction, urine pH, and stone risk.

RTA type	Primary defect	Serum K	Urine pH	High-yield associations	Stone/bone link
Type 1 distal	Cannot secrete H ⁺ in alpha-intercalated cells	Low	>5.5 despite acidosis	Autoimmune disease, amphotericin, Sjogren, sickle cell, obstruction	Calcium phosphate stones, nephrocalcinosis, bone buffering/osteomalacia
Type 2 proximal	Cannot reclaim filtered bicarbonate in proximal tubule	Low	Initially high, later can be <5.5 after bicarbonate falls	Fanconi syndrome, carbonic anhydrase inhibitors, myeloma, ifosfamide, tenofovir	Bone disease from phosphate wasting; less classic stone risk than type 1.
Type 4	Hypoaldosteronism or aldosterone resistance	High	Usually <5.5	Diabetes, CKD, ACEi/ARB, spironolactone, heparin, trimethoprim, NSAIDs	Hyperkalemia suppresses ammoniogenesis; stones less central.

A normal anion gap acidosis with hypokalemia should push you toward diarrhea, distal RTA, or proximal RTA. A normal anion gap acidosis with hyperkalemia should push you toward type 4 RTA, renal failure, or hypoaldosterone states. Urine pH helps most when interpreted after confirming systemic acidosis and excluding UTI with urease-producing organisms.

Nephrolithiasis: stone chemistry and acute stone care

Nephrolithiasis is a crystallization problem driven by supersaturation, low urine volume, urine pH, inhibitors of crystallization, and the concentration of calcium, oxalate, phosphate, uric acid, cystine, ammonium, and magnesium. The most common stones are calcium oxalate, followed by calcium phosphate, uric acid, struvite, and cystine. The immediate clinical syndrome is renal colic: severe flank pain radiating to the groin, hematuria, nausea/vomiting, and restlessness. The dangerous variant is an infected obstructed system, which is a urologic emergency because antibiotics alone may not penetrate a pressurized, obstructed collecting system adequately.



Initial management of uncomplicated renal colic prioritizes analgesia, antiemetics, assessment of kidney function, urinalysis, and imaging. NSAIDs are often excellent for renal colic because they reduce prostaglandin-mediated ureteral tone and renal pelvic pressure, but avoid or use caution in AKI, advanced CKD, severe volume depletion, high GI bleed risk, or pregnancy. Alpha-blockers such as tamsulosin may be used as medical expulsive therapy in selected distal ureteral stones. Urology involvement is urgent for infection plus obstruction, anuria, solitary kidney obstruction, uncontrolled pain/vomiting, worsening AKI, or large/proximal stones unlikely to pass.

Stone-size anchors: stones ≤5 mm often pass spontaneously; 5-10 mm stones have intermediate passage; stones >10 mm usually need intervention. Renal stones >2 cm are often managed with percutaneous nephrolithotomy, while ureteral stones may require ureteroscopy, lithotripsy, or stenting depending on location, size, anatomy, and infection status.

Obstructive uropathy and hydronephrosis

Obstruction raises pressure upstream of the blockage. If pressure is transmitted to Bowman space, the net filtration gradient falls and GFR declines. Unilateral obstruction may have little effect on serum creatinine when the opposite kidney is healthy. Bilateral ureteral obstruction, bladder outlet obstruction, obstruction in a solitary kidney, or obstruction superimposed on CKD can produce rapid AKI. Hydronephrosis on ultrasound supports obstruction, but early obstruction, dehydration, retroperitoneal fibrosis, and some intrarenal collecting system processes can complicate interpretation.

Level of obstruction	Common causes	Clinical clues	First moves
Intrarenal/tubular	Crystals, uric acid tumor lysis, acyclovir/indinavir crystals, myeloma casts	AKI with crystal risk; may not show classic hydronephrosis	Stop offender, increase tubular flow if appropriate, treat underlying cause.
Ureter	Stone, clot, tumor, stricture, retroperitoneal fibrosis	Colicky flank pain, hematuria, hydronephrosis	Analgesia, imaging, urology if infected/AKI/solitary kidney/uncontrolled symptoms.
Bladder outlet	BPH, prostate cancer, urethral stricture, neurogenic bladder, medications	Hesitancy, weak stream, retention, distended bladder, overflow incontinence	Bladder scan and Foley decompression; monitor post-obstructive diuresis.
External compression	Pregnancy, pelvic tumor, lymphadenopathy, fibrosis	Often subacute; bilateral if pelvic/retroperitoneal	Ultrasound/CT; stent or nephrostomy if kidney threatened.

Post-obstructive diuresis is common after relief of significant obstruction. It may be physiologic excretion of retained salt and water or pathologic tubular concentrating dysfunction. Monitor urine output, volume status, sodium, potassium, magnesium, phosphate, bicarbonate, and creatinine. Replace only enough fluid to prevent hypovolemia; matching 100% of urine output can perpetuate diuresis.

Urinary tract infection and pyelonephritis in the renal framework

Lower UTI is infection limited to the bladder and urethra; pyelonephritis is infection involving renal parenchyma and the collecting system. Cystitis causes dysuria, frequency, urgency, suprapubic discomfort, and pyuria. Pyelonephritis adds fever, chills, flank pain, costovertebral angle tenderness, nausea/vomiting, systemic illness, and sometimes AKI. WBC casts indicate inflammation within the kidney rather than just the bladder, but they are not exclusive to infection because AIN can also produce WBC casts.

Complicated UTI includes infection with functional/anatomic urinary tract abnormality, obstruction, catheter, renal transplant, pregnancy, male sex in many frameworks, immunocompromise, resistant organisms, or systemic illness. The management pivot is whether there is obstruction, sepsis, abscess, emphysematous infection, renal papillary necrosis, or failure to improve. Culture before antibiotics when possible, start empiric therapy based on severity and local resistance, then narrow. In an obstructed infected system, drainage is source control, not an optional add-on.

Scenario	Likely diagnosis	Testing	Management anchor
Dysuria/frequency without fever or flank pain	Acute cystitis	UA; culture if recurrent, complicated, pregnancy, male, resistant risk	Short-course oral antibiotic matched to local resistance and patient factors.
Fever, flank pain, pyuria, nausea/vomiting	Pyelonephritis	UA + urine culture; blood cultures if septic; imaging if severe or not improving	Antibiotics with renal dosing; IV if toxic, vomiting, pregnant, or complicated.
Fever + stone/hydronephrosis or anuria	Infected obstruction	Urgent imaging and cultures	Immediate urologic drainage plus antibiotics.

Scenario	Likely diagnosis	Testing	Management anchor
Diabetes + gas in renal parenchyma/collecting system	Emphysematous infection	CT	Broad antibiotics, glycemic control, drainage/nephrectomy depending severity.

Polycystic kidney disease and cystic renal disorders

Autosomal dominant polycystic kidney disease is a systemic cystic disorder, most often due to PKD1 or PKD2 variants. Cysts enlarge over decades, distort renal architecture, activate RAAS, and eventually reduce nephron function. Patients may present with hypertension, flank pain, hematuria, nephrolithiasis, recurrent cyst infection or UTI, enlarged kidneys, early satiety from mass effect, or progressive CKD. Extrarenal associations include hepatic cysts, intracranial aneurysms, mitral valve prolapse, abdominal wall hernias, and colonic diverticular disease.

The practical approach is blood pressure control, CKD risk reduction, avoidance of nephrotoxins, treatment of infection or stones, family counseling, and selective screening for intracranial aneurysm when there is a family history of aneurysm/subarachnoid hemorrhage, prior aneurysm, high-risk occupation, or concerning neurologic symptoms. Tolvaptan may slow kidney volume growth and decline in selected patients with rapidly progressive disease but requires monitoring for hepatotoxicity and counseling about aquaresis.

Autosomal recessive polycystic kidney disease usually presents in infancy or childhood with enlarged echogenic kidneys and congenital hepatic fibrosis. Acquired cystic kidney disease occurs in long-standing dialysis and increases renal cell carcinoma risk; it should not be confused with inherited ADPKD.

Selected GU disorders that intersect internal medicine

Renal/GU sections in medicine resources often include lower urinary tract and male GU disorders because they mimic renal disease, cause obstruction, or trigger infection. The internal medicine approach is to recognize emergencies and identify when urology is needed.

Disorder	Clinical pattern	Key danger	Management cue
Benign prostatic hyperplasia	Older man with hesitancy, weak stream, nocturia, retention, overflow symptoms	Postrenal AKI, recurrent UTI, bladder stones	Alpha-blocker for symptoms; 5-alpha-reductase inhibitor for large prostate; catheterize retention.
Prostatitis	Pelvic/perineal pain, urinary symptoms, fever; tender prostate in acute bacterial prostatitis	Sepsis, urinary retention; avoid vigorous prostate massage	Antibiotics with prostate penetration; longer course for chronic bacterial prostatitis.
Bladder cancer	Painless gross hematuria, smoking/chemical exposure	Delayed malignancy diagnosis	Cystoscopy and imaging; do not dismiss hematuria as UTI without follow-up.
Testicular torsion	Acute unilateral testicular pain, high-riding testis, absent cremasteric reflex	Ischemic loss of testis within hours	Immediate urologic exploration; do not delay for imaging if suspicion high.
Epididymitis	Gradual scrotal pain, posterior tenderness, urinary/STI risk	Abscess, infertility if undertreated	Antibiotics based on STI vs enteric organism risk; ultrasound if torsion uncertain.

Diagnostic algorithms

Algorithm 1: AKI with abnormal sediment

Start with hemodynamics, volume status, medications, obstruction screen, and baseline creatinine. Then let urine sediment and chemistry refine the differential.

Finding	Think first	Then check
Bland urine, hyaline casts, low urine sodium	Prerenal physiology	Volume loss, CHF/cirrhosis, NSAID/ACEi/ARB effect, sepsis vasodilation.
Muddy brown granular casts	ATN	Ischemia, sepsis, contrast risk, aminoglycosides, pigment injury.
WBC casts or sterile pyuria	AIN or pyelonephritis	Drug timeline, fever/rash/eosinophilia, urine culture, imaging if severe.
RBC casts or dysmorphic RBCs	Glomerulonephritis	Complements, ANA, ANCA, anti-GBM, hepatitis/HIV, biopsy logic.
Hydronephrosis or retention	Postrenal AKI	Foley, CT/ultrasound, urology, monitor post-obstructive diuresis.

Algorithm 2: Flank pain

Flank pain plus hematuria suggests stone, but fever, hypotension, pyuria, anuria, solitary kidney, pregnancy, transplant kidney, or rising creatinine changes the problem from pain control to urgent source control and kidney preservation. In stable uncomplicated suspected stone, noncontrast CT is highly sensitive; ultrasound is preferred in pregnancy and often used first when radiation reduction is important. If obstruction and infection coexist, drainage with ureteral stent or percutaneous nephrostomy is urgent.

Complications and emergency thresholds

Complication	Why it happens	Trigger for urgent action
Hyperkalemia	Reduced distal flow/secretion, acidosis, tissue breakdown, medications	ECG changes, K usually ≥ 6.5 , rapid rise, weakness, ongoing tissue breakdown, renal failure.
Metabolic acidosis	Reduced acid excretion, lactic acidosis from shock/sepsis, RTA	Severe acidemia, hemodynamic compromise, refractory acidosis, toxic alcohol concern.
Pulmonary edema	Oliguria/anuria plus fluid administration or heart failure	Hypoxemia, respiratory distress, refractory volume overload; dialysis if diuretics fail.
Infected obstruction	Bacteria behind a blocked collecting system	Fever/sepsis plus hydronephrosis/stone: urgent drainage.
Post-obstructive diuresis	Tubular dysfunction and excretion of retained solute/water after relief	High urine output with falling BP, rising Na, hypokalemia/hypomagnesemia.
Uremic complications	Accumulated toxins and inflammation in severe kidney failure	Pericarditis, encephalopathy, bleeding, seizures, severe nausea/vomiting.

The dialysis mnemonic AEIOU remains a useful emergency checklist: Acidosis, Electrolytes, Intoxications, Overload, and Uremic symptoms. Creatinine alone is not an absolute dialysis indication; the decision is driven by life-threatening complications, trajectory, and clinical context.

High-yield exam clues

- Muddy brown casts after hypotension or aminoglycosides -> ATN.
- Fever, rash, eosinophilia, sterile pyuria after beta-lactam/PPI/NSAID -> AIN.
- Dipstick blood with few RBCs after crush injury or seizure -> myoglobinuria from rhabdomyolysis.
- Normal anion gap acidosis + hypokalemia + urine pH > 5.5 -> distal RTA; think stones/nephrocalcinosis.
- Normal anion gap acidosis + hyperkalemia in diabetes/CKD/ACEi/spironolactone -> type 4 RTA.
- Hexagonal crystals -> cystine stone; coffin-lid crystals/staghorn -> struvite infection stone; radiolucent stone with acidic urine -> uric acid.
- Fever plus obstructing stone -> antibiotics plus urgent drainage, not antibiotics alone.
- Painless gross hematuria in smoker -> evaluate for bladder cancer even if symptoms temporarily improve.
- Acute scrotal pain with absent cremasteric reflex -> torsion until proven otherwise.

Common pitfalls

- Calling every AKI with edema "intrinsic renal failure." Heart failure and cirrhosis can cause avid sodium retention and prerenal physiology despite total-body fluid excess.
- Overtrusting FeNa after diuretics. In that setting, sediment, FeUrea, clinical timeline, and response to therapy are more informative.
- Missing obstruction because only one kidney is blocked. Serum creatinine may remain near baseline with unilateral obstruction if the other kidney works.
- Treating infected obstructed stone with antibiotics alone. Drainage is source control.
- Giving NSAIDs for renal colic to a patient with active AKI, advanced CKD, or severe volume depletion without weighing risk.
- Ignoring post-obstructive diuresis after Foley placement in severe retention.

- Assuming pyuria always means infection. AIN, stones, TB, STIs, and inflammatory disorders can cause sterile pyuria.
- Assuming creatinine level alone determines dialysis. Dialysis is driven by refractory complications and symptoms.

Practice questions

Q1. A 72-year-old man is admitted with septic shock. Two days later, creatinine rises from 0.9 to 2.8 mg/dL. Urine microscopy shows muddy brown granular casts. FeNa is 3%. What is the most likely renal process?

Answer: Ischemic acute tubular necrosis. The timing after shock, granular casts, impaired sodium reclamation, and intrinsic AKI chemistry support ATN.

Q2. A 49-year-old woman develops AKI 10 days after starting a proton pump inhibitor. She has sterile pyuria and WBC casts but no edema and only mild proteinuria. What is the first management step?

Answer: Stop the offending medication. This is drug-induced acute interstitial nephritis until proven otherwise; biopsy/nephrology is considered if kidney function fails to improve or diagnosis is uncertain.

Q3. A patient after a generalized seizure has dark urine. Dipstick is strongly positive for blood, but microscopy shows 0-2 RBC/hpf. Potassium is elevated and phosphate is elevated. What diagnosis should be considered?

Answer: Rhabdomyolysis with myoglobinuria causing pigment nephropathy risk.

Q4. A patient has normal anion gap metabolic acidosis, hypokalemia, nephrocalcinosis, and urine pH 6.2 despite systemic acidosis. What is the diagnosis?

Answer: Distal/type 1 renal tubular acidosis.

Q5. A febrile patient has flank pain, pyuria, rising creatinine, and CT shows an obstructing ureteral stone with hydronephrosis. What is the key intervention in addition to antibiotics?

Answer: Urgent decompression of the collecting system by ureteral stent or percutaneous nephrostomy.

Q6. After Foley placement for urinary retention, a patient produces 5 L of urine in the next 12 hours and becomes dizzy. What complication is occurring?

Answer: Post-obstructive diuresis with risk of hypovolemia and electrolyte loss.

Q7. A recurrent stone former has radiolucent stones and urine pH 5.0. What stone type and prevention strategy fit best?

Answer: Uric acid stones; urine alkalinization with potassium citrate and reduction of uric acid burden when indicated.

Q8. A young man has sudden severe unilateral testicular pain and absent cremasteric reflex. What is the management priority?

Answer: Immediate urologic evaluation/exploration for torsion; do not delay for testing if suspicion is high.

Graph-RAG extraction map

The following concept map is designed for Medvale graph extraction. Nodes should preserve disease location, mechanism, urine signature, complications, and management triggers. Edges should explain the relationship, not only connect labels.

Node	Class / subclass	Key properties	High-value edges
Acute tubular necrosis	Renal disease -> intrinsic AKI -> tubular injury	Ischemic or nephrotoxic; muddy brown casts; FeNa often >2%; recovery diuresis	Shock -> ATN; aminoglycosides -> ATN; ATN -> hyperkalemia/acidosis; ATN -> dialysis indication if complications.
Acute interstitial nephritis	Renal disease -> tubulointerstitial inflammation	Drug/infection/systemic triggers; sterile pyuria; WBC casts; mild proteinuria	PPI/beta-lactam/NSAID -> AIN; AIN mimics pyelonephritis; drug withdrawal treats AIN.

Node	Class / subclass	Key properties	High-value edges
Rhabdomyolysis	Muscle injury -> pigment nephropathy risk	CK elevation; myoglobinuria; hyperkalemia; hypocalcemia early	Crush/seizure/statin -> rhabdo; rhabdo -> ATN; fluids prevent pigment AKI.
Renal tubular acidosis	Acid-base -> normal anion gap metabolic acidosis	Type 1 low K high urine pH; type 2 proximal bicarbonate wasting; type 4 hyperK	RTA -> nephrolithiasis; diabetes -> type 4 RTA; Sjogren -> distal RTA.
Nephrolithiasis	GU/renal -> crystal disease	Stone chemistry; colic; hematuria; obstruction/infection risk	Low urine volume -> stones; urease organisms -> struvite; obstructing stone + infection -> urgent drainage.
Obstructive uropathy	Renal disease -> postrenal AKI	Hydronephrosis; retention; bilateral/solitary kidney causes AKI	BPH -> obstruction; obstruction -> hydronephrosis; decompression -> post-obstructive diuresis.
Pyelonephritis	Infectious disease -> upper UTI	Fever, flank pain, pyuria, WBC casts, culture	Ascending infection -> pyelonephritis; obstruction worsens infection; pyelonephritis mimics AIN.
ADPKD	Inherited renal disease -> cystic kidney disease	Bilateral cysts; HTN; hematuria; stones; CKD; extrarenal aneurysm/hepatic cysts	PKD -> CKD; ADPKD -> intracranial aneurysm; cyst infection -> fever/flank pain.

References used for clinical cross-checking

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